Feature Selection and PCA

Choosing What Matters in Machine Learning

Core idea

- Feature selection keeps the most useful input features.
- It can reduce complexity and improve model efficiency.
- PCA reduces dimensionality by transforming features.



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Three Ways to Select Features

Feature selection methods specified in the assessment question

1. Filter Methods

- Use statistical measures before model training.
- Fast and simple.
- Example: correlation or information-based ranking.

2. Wrapper Methods

- Test different feature subsets with a model.
- Choose the subset giving better performance.
- More computationally expensive.

3. Embedded Methods

- Selection happens during model training.
- Combines selection with learning.
- Example: tree-based feature importance.

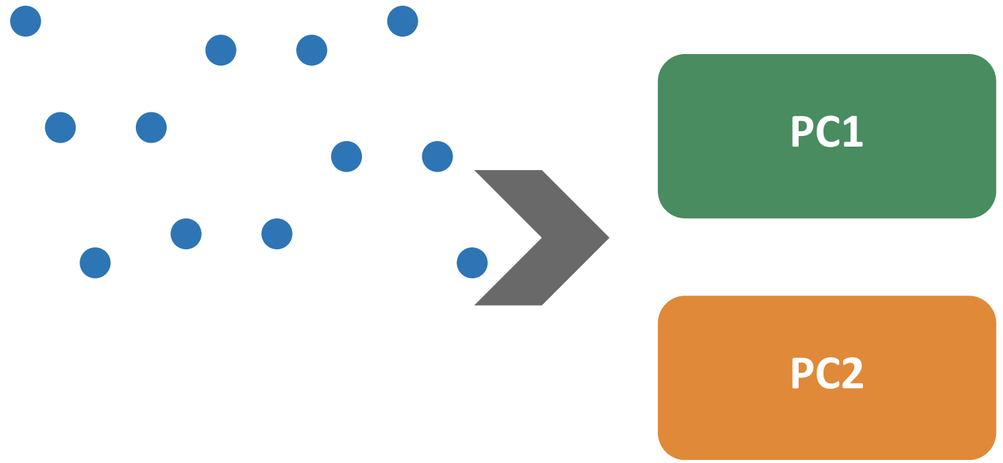
PCA: Dimensionality Reduction

Principal Component Analysis transforms many correlated features into fewer components

How PCA works

- Find directions that capture the most variation in the data.
- Create new variables called principal components.
- Keep the most informative components and discard less useful ones.
- Result: fewer dimensions while retaining important information.

$$PC1=w1X1+w2X2+w3X3+w4X4+w5X5$$
$$PC2=V1X1+V2X2+V3X3+V4X4+V5X5$$



Many original features → fewer principal components

Simple Application: Student Performance Prediction

Example of how feature selection can simplify an ML problem

Original features	Selected features	Benefit
Attendance	Attendance	Useful signal
Study hours	Study hours	Useful signal
Sleep hours	Sleep hours	Useful signal
Phone usage	—	Removed if less useful
Social media time	—	Removed if redundant

Key takeaway: Keep informative features, reduce unnecessary dimensions, and make ML models simpler and more efficient.